Genetic diseases – mutations

Sometimes the codes in genes are changed, and the developing cells receive a wrong message. A change in the codes could cause the cells to make a wrong protein, or no protein at all. As a result, the cells do not function normally. Changes in the genes are called **mutations**, derived from a Latin word meaning *change*. Mutations can be dominant or recessive. If dominant, they will be expressed and hide the normal form of the gene. If recessive, the normal gene will hide the mutation. To have the characteristic of the recessive mutation, a person must have both genes with that mutation.

The table below summarises some mutations or genetic disorders in humans.

Name of disorder	Description	Frequency	Dominance and position
Albinism	• White skin, white hair, pink eyes, poor vision because unable to produce pigment	1 in 20 000	Recessive On eleventh chromosome
Cystic Fibrosis	• Usually lethal • Breathing difficult because mucus in bronchial tubes thick and restricting • Digestion difficult because mucus in pancreatic ducts thick and blocking	1 in 2 500 Caucasians	Recessive On second chromosome
Duchenne Muscular Dystrophy	• Waddling gait, toe walking • Frequent falls, difficulty in getting up • Confined to wheelchair • Death usually at age of about 20 years because of weak and wasted muscles • Males are rarely fathers, but females can be carriers without showing disease but passing on faulty gene	1 in 5 000 males	Recessive Sex linked, on X chromosome
Haemophilia	• Can be lethal • Can bleed to death from minor injury • Internal bleeding particularly around joints • Blood does not clot because clotting agent is absent, so bleeding does not stop	1 in 5 000 males	Recessive Sex linked, on X chromosome
Huntington Disease	• Neurological disorder • Lethal 10 to 15 years after start of symptoms • Normal until middle age, then muscular spasms and personality disorders because of progressive degeneration of brain cells • Affected parents may have affected children before symptoms show	1 in 20 000	Dominant On fourth chromosome
Phenylketouria (PKU)	• Metabolic disorder • Severe mental retardation because enzyme is missing that breaks down amino acid phenylalamine resulting in poisoning cells • Newborns are placed on diet low in phenylalamine until about 7 years, when brain is fully developed	1 in 5 000 caucasians	Recessive On twelfth chromosome
Tourette Syndrome	• A tic disorder: 'twitches and noises' • On average starts at 7 years • Involuntary movements: eye blinking, nose twitching, making grimaces, arm and leg twitches • Involuntary noises: grunting, throat clearing, shouting, saying obscenities, barking • Often only mild symptoms	1 in 1 500 more common in males than females	Dominant On nineteenth chromosome ? Possibly on other chromosomes

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1 Write a report. Select one genetic disorder and write a report about it. There should be a short introductory paragraph and then several ordered information paragraphs. Use the scaffold below to help you organise your paragraphs. You could also use the library or Internet to research extra information about the disorder.

- **Introduction:** What is the name of the disorder you have chosen?
- **Description 1:** Describe the disorder, using complete sentences.
- **Description 2:** What is the frequency of the disorder? Is it a recessive or dominant disorder? What chromosomes are affected?
- **Conclusion:** Summarise the disease in one sentence.

2 Now select a different genetic disorder. Write a report about it, following the format described in question 1.

3 Solve the clues to complete the crossword puzzle. The sentence in the middle of the puzzle is about a person who was a famous musician and who was considered rude and uncontrollable. Geneticists believe he may have suffered from this disease. Ignore the spaces between words in this sentence.

Clues

a Changes in genes are called _____.

b Mutations give cells the _____ message.

c This is needed to break down the amino acid phenylalamine.

d A person who appears pale and has pink-coloured eyes may have _____.

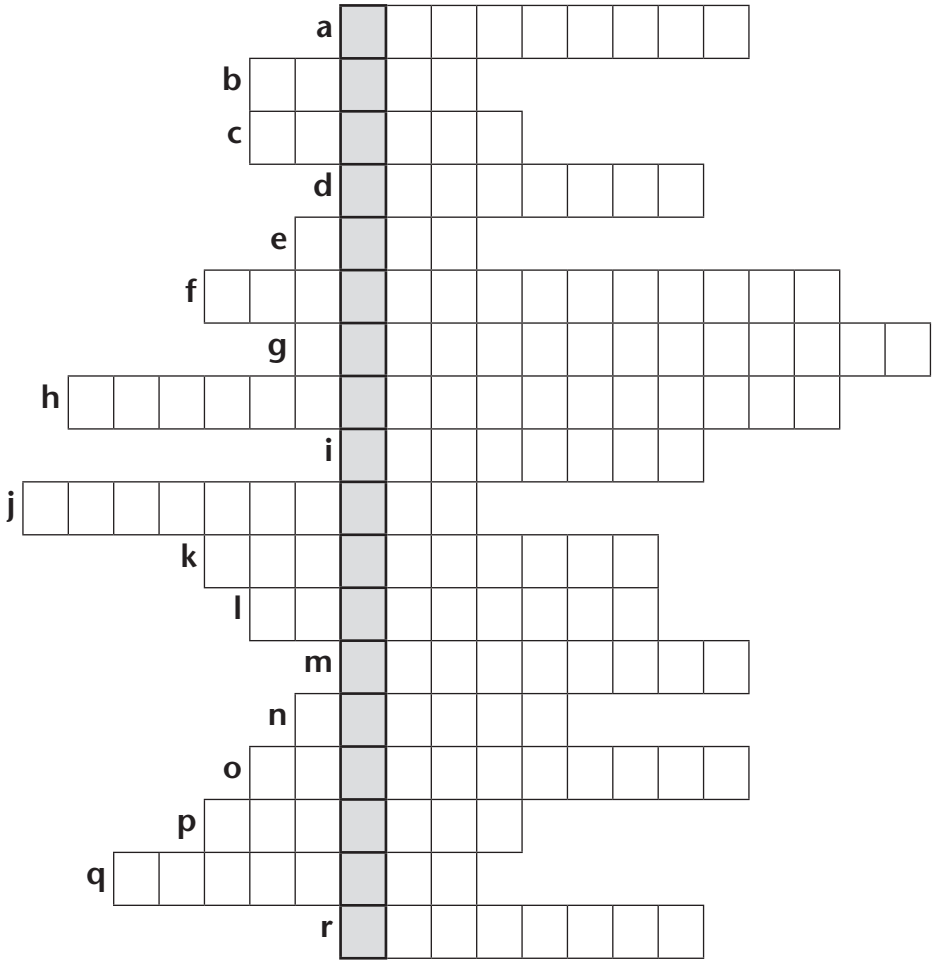
e Huntington disease is located on chromosome 4. True or false?

f Genetic disorder that causes thick mucus to accumulate in pipes in the lungs and pancreas.

g The abbreviation for this disease is PKU.

h When children find walking and getting up difficult, they may suffer from _____.

i If mutations are expressed with only one gene, they are said to be _____.



j A disorder that appears later in life and involves severe muscle spasms.

k A sex-linked disease where people lack a blood clotting agent; mainly found in males.

l Group of people with a long face, medium-to-tall stature, light-coloured skin, originating from Europe.

m People must have two genes for a _____ characteristic to appear.

n Deadly

o If newborns have PKU, mental _____ can be avoided if they eat a special diet.

p Genes code for _____.

q Organ that helps in digestion.

r Descriptions of a disease can also be referred to as the _____ of the disease.

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